

EXAM 7 – SPRING 2015

20. (3 points)

Given the following information for an insurer's gross aggregate loss distribution:

Expected value	\$20,000,000
90% VaR	\$50,000,000
Limited expected value at \$50,000,000	\$8,000,000

- The capital requirement is set at 90% TVaR.

The insurer is evaluating the impact of two reinsurance options from a reinsurer that has received a B credit rating from a widely recognized rating agency. Assume a 60% charge for reinsurance recoverable credit risk associated with this reinsurer.

Option 1

- Excess-of-loss reinsurance
- Attachment \$50,000,000
- Limit \$100,000,000
- Expected ceded loss \$7,500,000

Option 2

- Quota share reinsurance
- 40% ceding percentage
- No ceding commission

a. (1.25 points)

Calculate the net capital requirement for the insurer if Option 1 is purchased.

b. (0.75 point)

Calculate the net capital requirement for the insurer if Option 2 is purchased.

c. (0.5 point)

Briefly describe two challenges of assessing the impact of reinsurance on the insurer's risk profile.

d. (0.5 point)

Briefly describe two ways the insurer may be able to lower its reinsurance recoverable credit risk exposure.

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EXAM 7 SPRING 2015 SAMPLE ANSWERS AND EXAMINERS' REPORT

QUESTION 20	
TOTAL POINT VALUE: 3	LEARNING OBJECTIVE: C1: Demonstrate how insurance and financial risk can be analyzed quantitatively. C4: Demonstrate the properties of various risk measures and their limitations. C8: Evaluate best practices in risk measurement, modeling, and management of various financial and non-financial risks faced by an entity
SAMPLE ANSWERS	
Part a: 1.25 points	
<i>Sample Answer 1</i>	
$TVaR_{\alpha}(X_{Gross}) = VaR_{\alpha}(X_{Gross}) + \frac{E[X_{Gross}] - E[X_{Gross} \wedge VaR_{\alpha}(X_{Gross})]}{1 - \alpha} = 50M + \frac{20M - 8M}{1 - 0.9} = 170M$ $TVaR_{\alpha}(X_{Cede}) = VaR_{\alpha}(X_{Cede}) + \frac{E[X_{Cede}] - E[X_{Cede} \wedge VaR_{\alpha}(X_{Cede})]}{1 - \alpha} = \frac{7.5M}{1 - \alpha} = 75M$ Net capital requirement $= TVaR_{\alpha}(X_{Gross}) - (1 - \theta) * TVaR_{\alpha}(X_{Cede}) = 170M - (1 - 0.6) * 75M = 140M$	
<i>Sample Answer 2</i>	
$TVaR_{\alpha}(X_{Net}) = VaR_{\alpha}(X_{Net}) + \frac{E[X_{Net}] - E[X_{Net} \wedge VaR_{\alpha}(X_{Net})]}{1 - \alpha} = 50M + \frac{20M - 7.5M - 8M}{1 - 0.9} = 95M$ $TVaR_{\alpha}(X_{Cede}) = VaR_{\alpha}(X_{Cede}) + \frac{E[X_{Cede}] - E[X_{Cede} \wedge VaR_{\alpha}(X_{Cede})]}{1 - \alpha} = \frac{7.5M}{1 - \alpha} = 75M$ Net capital requirement $= TVaR_{\alpha}(X_{Net}) + \theta * TVaR_{\alpha}(X_{Cede}) = 95M + 0.6 * 75M = 140M$	
<i>Sample Answer 3</i>	
$TVaR_{\alpha}(X_{Gross}) = VaR_{\alpha}(X_{Gross}) + \frac{E[X_{Gross}] - E[X_{Gross} \wedge VaR_{\alpha}(X_{Gross})]}{1 - \alpha} = 50M + \frac{20M - 8M}{1 - 0.9} = 170M$ $TVaR_{\alpha}(X_{Net}) = VaR_{\alpha}(X_{Net}) + \frac{E[X_{Net}] - E[X_{Net} \wedge VaR_{\alpha}(X_{Net})]}{1 - \alpha} = 50M + \frac{20M - 7.5M - 8M}{1 - 0.9} = 95M$ $TVaR_{\alpha}(X_{Cede}) = TVaR_{\alpha}(X_{Gross}) - TVaR_{\alpha}(X_{Net}) = 170M - 95M = 75M$ Net capital requirement $= TVaR_{\alpha}(X_{Gross}) - (1 - \theta) * TVaR_{\alpha}(X_{Cede}) = 170M - (1 - 0.6) * 75M = 140M$	
Part b: 0.75 point	
<i>Sample Answer 1</i>	

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$$TVaR_{\alpha}(X_{Gross}) = VaR_{\alpha}(X_{Gross}) + \frac{E[X_{Gross}] - E[X_{Gross} \wedge VaR_{\alpha}(X_{Gross})]}{1 - \alpha} = 50M + \frac{20M - 8M}{1 - 0.9} = 170M$$

$$TVaR_{\alpha}(X_{Cede}) = 0.4 * TVaR_{\alpha}(X_{Gross}) = 0.4 * 170M = 68M$$

Net capital requirement

$$= TVaR_{\alpha}(X_{Gross}) - (1 - \theta) * TVaR_{\alpha}(X_{Cede}) = 170M - (1 - 0.6) * 68M = 142.8M$$

Sample Answer 2

Note on other accepted answers:

If Gross TVaR was miscalculated in Part A (e.g. 120 instead of 170), then 120 was treated as correct in Part B and full credit was given in Part B for answers based on that 120.

Part c: 0.5 point

Sample Answer 1

The complexity of reinsurance contracts makes it harder to calculate the impact of loss. XS loss does not have a linear relationship with the underlying risks.

Sample Answer 2

Some reinsurance contracts do not warrant a credit to the insurer's risk – likely done for tax reasons.

Part d: 0.5 point

Sample Answer 1

Reduce concentration risk – diversify its reinsurance contract with many reinsurers in many different regions.

Sample Answer 2

Choose reins. with good credit ratings to reduce default risk.

EXAMINERS' REPORT

- Candidate were expected to know terms of insurance and financial risk, know the properties of various risk measures and their limitations (e.g. TVaR), and know best practices in measurement, modeling, and managing of risk.
- Candidates generally struggled on Parts a and b – application of risk measures – and scored well on Parts c and d – conceptual understanding of managing insurance and financial risk.
- The individual calculations of TVaR were straightforward, but the application of those individual TVaR calculations into the net capital requirement (based on the specifics of the reinsurance contracts) was a challenge for many candidates.
 - Some candidates misapplied the 90% factor for TVaR, by either multiplying by 0.9 or dividing by 0.9, instead of dividing by (1-0.9)
 - There was also some misunderstanding on the relationship of net, ceded, and gross, and how a recoverable credit risk charge should be applied.

Part a

Candidates struggled with this question. It was common that the candidates did not know the formulas used to calculate TVaR and the Net Capital Requirement. It was common that the candidates did not enter the proper values into the calculation of TVaR and the Net Capital

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Requirement.
Part b
Same comments as for part a above.
Part c
Candidates generally did well on this question.
Part d
Candidates generally did well on this question.